Teaching notes

Suggested starter activity

Establish that bones need to be strong to support the weight of our body by discussing the size and shape of the femurs.

Demonstrate the need for their shape by showing that a rolled paper tower will support a pile of books.

Show that bones need to be flexible by hanging a weight on a straw and a thick, brittle length of spaghetti. Stress that brittle bones break easily and ask if anyone knows about brittle bone disease.

Main activity

NB: A risk assessment should be carried out before this practical is used with students. See CLEAPPS (www.cleapss.org.uk) for further information. When heating the bone goggles should be worn at all times as they may shatter. As an alternative bones could be heated in an oven prior to the lesson. If using chicken bones which may be contaminated with salmonella ask students to wash their hands thoroughly after handling them.

Students should work in groups of 2 or 3.

Each group will need:

- worksheet
- 2 bone samples
- Bunsen burner
- heatproof mat
- splint
- goggles
- metal tongs
- beaker

Ask students to note down their initial observations of the bone. Then discuss the second section of worksheet. Stress need for careful handling of bone in flame as too tight a squeeze will shatter the hot bone! Goggles must be worn at all time and hot bone must be left on the mat.

Further activities

Research and write notes on brittle bone disease.

Bone is a composite material. It needs to be strong but flexible. Its strength comes from the mineral, calcium phosphate, and its flexibility from an organic material, collagen.

Your task: to investigate the two factors in bone and their effect on the composite material by removing each material from bone.

Task 1 – initial observation of bone

- Look at your piece of bone and describe it in words.
- Draw your sample labelling any features that you notice.
- Feel the bone and describe its texture and flexibility.

Task 2 – removing the collagen

You can roast the collagen out of the bone.

- Hold the fresh piece of bone in tongs and place in the hottest part of the Bunsen flame.
- Move the bone about in the flame until the bone becomes blackened.
- Carefully place the very hot bone onto a heatproof mat and leave it to cool.
- Write up your investigation so far; include a labelled diagram to show how you heated the bone.
- When it has cooled down handle the bone and compare it with your initial observations.

Task 3 – removing the calcium phosphate

You can use a chemical reaction to remove this from bone.

- Put your piece of bone into a beaker so that it lies comfortably across the bottom.
- Carefully pour some vinegar or lemon juice to cover the bone.
- Label your beaker and place on the windowsill for two or three days.
- Write up your investigation so far; include a fully labelled diagram of the bone in the acid.

After 2 or 3 days:

- Have a beaker of water ready on your table.
- Collect your bone in its beaker and very carefully use tongs to remove it from the acid and rinse it thoroughly in the beaker of water.
- Place your rinsed bone onto a paper towel.
- Handle your rinsed bone and compare it to your initial observations.